# Block practice · Hexadecimal, shifts, characters, sound and RLE

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Hexadecimal: why we use it and how to convert

ASCII and Unicode character sets

Run-length encoding (RLE)

Logical and arithmetic binary shifts

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# Practice workbook · Hexadecimal: why we use it and how to convert

Why programmers use hexadecimal (base 16, one hex digit per nibble, shorter and less error-prone than binary) and converting between hexadecimal, binary and denary.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-04

# 1. Question 3 · part a.i · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part a.i only.

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**3** Computers use binary to represent and store data.

(a) Binary and hexadecimal can be used to represent numbers.

(i) Convert the binary pattern 1011 1001 to hexadecimal.

**(2)**

(ii) Identify the statement that is true for hexadecimal.

**(1)**

- ☒ **A** Hexadecimal can be processed faster by a computer than binary
- ☒ **B** Hexadecimal digits can be represented using one nibble
- ☒ **C** Hexadecimal uses less storage space in RAM than binary
- ☒ **D** Hexadecimal uses the number base 6

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(iii) Describe the binary shift that is equivalent to dividing an unsigned binary number by 4.

**(2)**

P 7 8 6 3 7 A 0 7 2 0

## 1. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

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## 2. Question 2 · part b · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b only.

**2** Computers use binary digits to represent data.

(a) Sign and magnitude and two's complement are two ways of representing signed integer numbers.

(i) Convert the denary number -94 to a binary pattern using sign and magnitude representation.

**(2)**

(ii) 1000 1001 is a signed integer that uses two's complement representation.

Convert this signed integer to denary.

**(2)**

(b) Denary numbers use base 10

Identify the base for hexadecimal numbers.

**(1)**

- ☒ **A** 32
- ☒ **B** 16
- ☒ **C** 8
- ☒ **D** 2

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## 2. Question 2 · part b · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b only.

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(c) Computers encode characters using ASCII and Unicode.

(i) State the number of characters that can be represented using standard ASCII.

**(1)**

(ii) The ASCII system uses 7 bits to represent a character. The ASCII code for the character 'A' using denary is 65; other alphabetical characters follow on from this in sequence.

Identify the ASCII code for 'H'.

**(1)**

- ☒ **A** 1000111
- ☒ **B** 1001010
- ☒ **C** 1001000
- ☒ **D** 1000101

(iii) Explain why Unicode was developed.

**(2)**

**(Total for Question 2 = 9 marks)**

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## 2. Question 2 · part b · 1 mark · Working and answer

Working and answer

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## 3. Question 1 · part a · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part a only.

**1** Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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## 3. Question 1 · part a · 1 mark · Working and answer

Working and answer

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## 4. Question 1 · part b · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

**1** Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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## 4. Question 1 · part b · 2 marks · Working and answer

Working and answer

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## 5. Question 1 · part a.iii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part a.iii only.

**1** Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

**(2)**

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

**(2)**

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(iii) Convert the binary pattern 11000110 to hexadecimal.

**(2)**

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DO NOT WRITE IN THIS AREA

## 5. Question 1 · part a.iii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part a.iii only.

(b) Images are stored as binary data.

(i) A bitmap image is 400 pixels wide by 200 pixels high.

It has a colour depth of 12 bits.

Construct an expression to show how the image size, in bytes, is calculated.

Do **not** calculate the answer.

**(2)**

(ii) **Figure 1** is an image of an icon, set out in an 8 × 8 pixel grid.

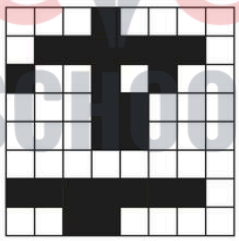


**Figure 1**

A run-length encoding compression algorithm uses:

- W for white pixels
- B for black pixels.

Give the result of compressing the first 4 lines of the icon using the algorithm.

**(3)**

Line 1 .....

Line 2 .....

Line 3 .....

Line 4 .....

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## 5. Question 1 · part a.iii · 2 marks · Working and answer

Working and answer

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## 6. Question 2 · part a · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

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**2** Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

**(2)**

| Denary | Hexadecimal |
|---|---|
| 8 | |
| 12 | |

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

**(1)**

- ☒ **A** $6^2$
- ☒ **B** $6 \times 2$
- ☒ **C** $6^2 - 1$
- ☒ **D** $2^6$

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

**(1)**

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

**(1)**

P 7 2 9 3 7 A 0 5 2 4

## 6. Question 2 · part a · 2 marks · Working and answer

Working and answer

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# 7. Question 3 · part a.ii · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part a.ii only.

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DO NOT WRITE IN THIS AREA

**3** Computers use binary to represent and store data.

(a) Binary and hexadecimal can be used to represent numbers.

(i) Convert the binary pattern 1011 1001 to hexadecimal.

**(2)**

(ii) Identify the statement that is true for hexadecimal.

**(1)**

- ☒ **A** Hexadecimal can be processed faster by a computer than binary
- ☒ **B** Hexadecimal digits can be represented using one nibble
- ☒ **C** Hexadecimal uses less storage space in RAM than binary
- ☒ **D** Hexadecimal uses the number base 6

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(iii) Describe the binary shift that is equivalent to dividing an unsigned binary number by 4.

**(2)**

P 7 8 6 3 7 A 0 7 2 0

## 7. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

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# Practice workbook · ASCII and Unicode character sets

How characters are encoded: ASCII bits and size, finding a code by counting on from a given code, benefits of ASCII, and why Unicode was needed.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-09

# 1. Question 2 · part a.iii · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part a.iii only.

**2** Computers use binary to represent and store data.

(a) The denary number 78 is the ASCII code for the character **N**.

(i) Convert the denary number 78 to 8-bit binary.

**(2)**

(ii) Identify the number of characters that can be represented using standard ASCII.

**(1)**

- ☒ **A** 64
- ☒ **B** 128
- ☒ **C** 256
- ☒ **D** 512

(iii) Explain **one** reason for using Unicode rather than ASCII to encode languages other than English.

**(2)**

(b) Convert the denary number –43 to 8-bit binary using sign and magnitude representation.

**(2)**

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P 7 2 5 3 8 A 0 4 2 0

## 1. Question 2 · part a.iii · 2 marks · Working and answer

Working and answer

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# 2. Question 2 · part c.i · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c.i only.

**2** Computers use binary digits to represent data.

(a) Sign and magnitude and two's complement are two ways of representing signed integer numbers.

(i) Convert the denary number -94 to a binary pattern using sign and magnitude representation.

**(2)**

(ii) 1000 1001 is a signed integer that uses two's complement representation.

Convert this signed integer to denary.

**(2)**

(b) Denary numbers use base 10

Identify the base for hexadecimal numbers.

**(1)**

- ☒ **A** 32
- ☒ **B** 16
- ☒ **C** 8
- ☒ **D** 2

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# 2. Question 2 · part c.i · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c.i only.

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(c) Computers encode characters using ASCII and Unicode.

(i) State the number of characters that can be represented using standard ASCII.

**(1)**

(ii) The ASCII system uses 7 bits to represent a character. The ASCII code for the character 'A' using denary is 65; other alphabetical characters follow on from this in sequence.

Identify the ASCII code for 'H'.

**(1)**

- ☒ **A** 1000111
- ☒ **B** 1001010
- ☒ **C** 1001000
- ☒ **D** 1000101

(iii) Explain why Unicode was developed.

**(2)**

**(Total for Question 2 = 9 marks)**

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## 2. Question 2 · part c.i · 1 mark · Working and answer

Working and answer

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## 3. Question 2 · part c.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c.ii only.

**2** Computers use binary digits to represent data.

(a) Sign and magnitude and two's complement are two ways of representing signed integer numbers.

(i) Convert the denary number -94 to a binary pattern using sign and magnitude representation.

**(2)**

(ii) 1000 1001 is a signed integer that uses two's complement representation.

Convert this signed integer to denary.

**(2)**

(b) Denary numbers use base 10

Identify the base for hexadecimal numbers.

**(1)**

- ☒ **A** 32
- ☒ **B** 16
- ☒ **C** 8
- ☒ **D** 2

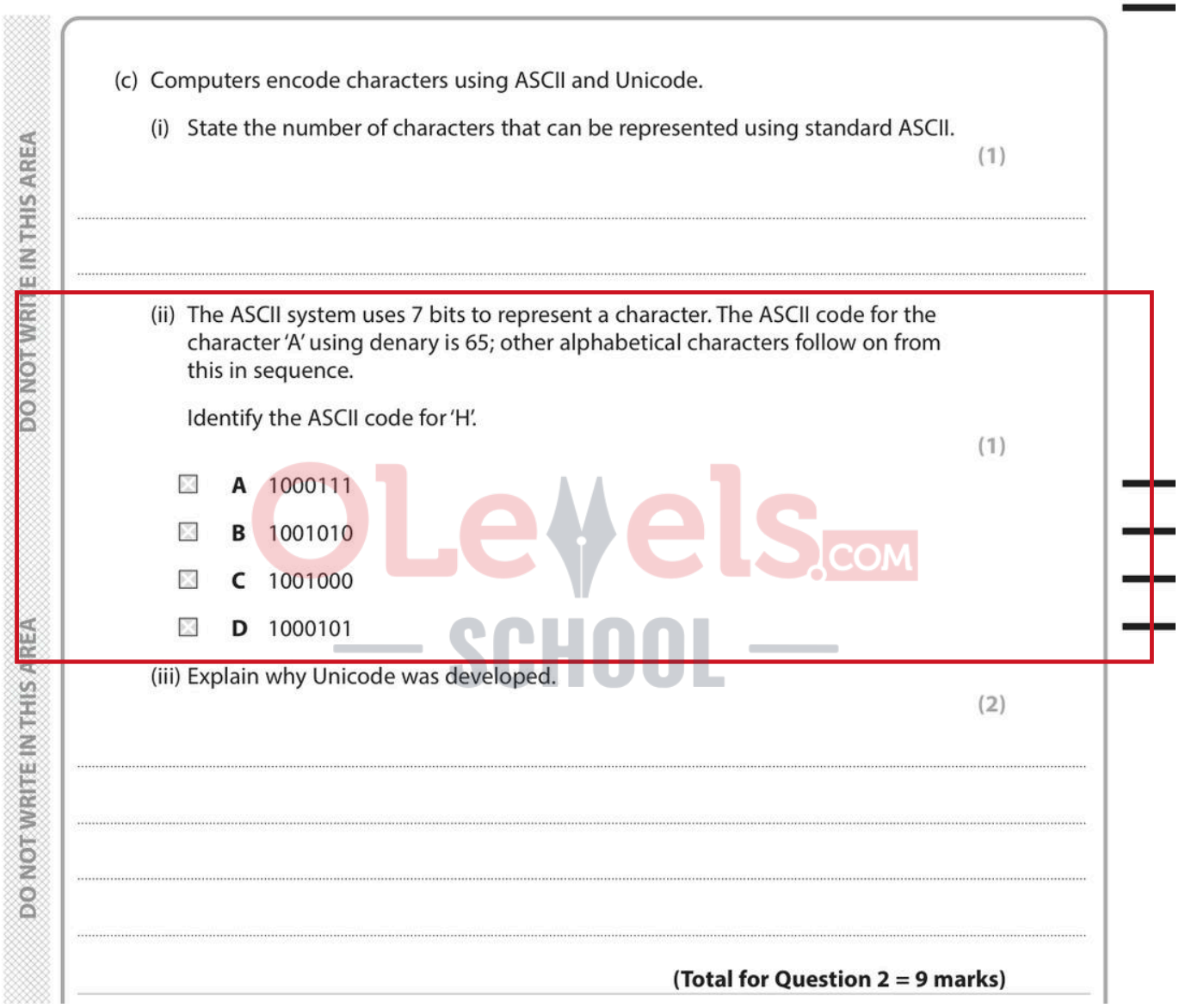
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## 3. Question 2 · part c.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c.ii only.



DO NOT WRITE IN THIS AREA

(c) Computers encode characters using ASCII and Unicode.

(i) State the number of characters that can be represented using standard ASCII.

(1)

(ii) The ASCII system uses 7 bits to represent a character. The ASCII code for the character 'A' using denary is 65; other alphabetical characters follow on from this in sequence.

Identify the ASCII code for 'H'.

(1)

- ☒ **A** 1000111
- ☒ **B** 1001010
- ☒ **C** 1001000
- ☒ **D** 1000101

(iii) Explain why Unicode was developed.

(2)

**(Total for Question 2 = 9 marks)**

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## 3. Question 2 · part c.ii · 1 mark · Working and answer

Working and answer

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# 4. Question 2 · part c.iii · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c.iii only.

**2** Computers use binary digits to represent data.

(a) Sign and magnitude and two's complement are two ways of representing signed integer numbers.

(i) Convert the denary number -94 to a binary pattern using sign and magnitude representation.

**(2)**

(ii) 1000 1001 is a signed integer that uses two's complement representation.

Convert this signed integer to denary.

**(2)**

(b) Denary numbers use base 10

Identify the base for hexadecimal numbers.

**(1)**

- ☒ **A** 32
- ☒ **B** 16
- ☒ **C** 8
- ☒ **D** 2

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# 4. Question 2 · part c.iii · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c.iii only.

DO NOT WRITE IN THIS AREA

(c) Computers encode characters using ASCII and Unicode.

(i) State the number of characters that can be represented using standard ASCII.

(1)

.....

.....

(ii) The ASCII system uses 7 bits to represent a character. The ASCII code for the character 'A' using denary is 65; other alphabetical characters follow on from this in sequence.

Identify the ASCII code for 'H'.

(1)

☒ **A** 1000111

☒ **B** 1001010

☒ **C** 1001000

☒ **D** 1000101

DO NOT WRITE IN THIS AREA

(iii) Explain why Unicode was developed.

(2)

.....

.....

.....

.....

**(Total for Question 2 = 9 marks)**

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## 4. Question 2 · part c.iii · 2 marks · Working and answer

Working and answer

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## 5. Question 1 · part f · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f only.

**1** Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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## 5. Question 1 · part f · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f only.

(d) Complete the table by adding these two 8-bit binary integers.

**(2)**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| | | | | | | | |

(e) Give the result of performing a logical left shift of 1 place on the binary integer 0100.

**(1)**

.....

(f) Give the number of bits per character used by standard ASCII.

**(1)**

.....

(g) **Figure 1** shows a black and white bitmap image.

The pixels in row 5 are represented by the binary pattern 1001 1001

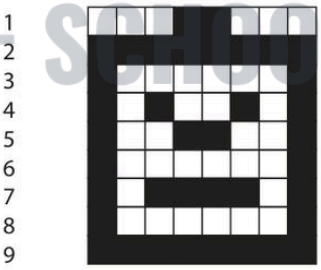



**Figure 1**

(i) State what is meant by the term **pixel**.

**(1)**

.....

(ii) Construct an expression to show the number of pixels in the image.

**(1)**

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## 5. Question 1 · part f · 1 mark · Working and answer

Working and answer

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# 6. Question 2 · part a.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part a.ii only.

**2** Computers use binary to represent and store data.

(a) The denary number 78 is the ASCII code for the character **N**.

(i) Convert the denary number 78 to 8-bit binary.

**(2)**

(ii) Identify the number of characters that can be represented using standard ASCII.

**(1)**

- ☒ **A** 64
- ☒ **B** 128
- ☒ **C** 256
- ☒ **D** 512

(iii) Explain **one** reason for using Unicode rather than ASCII to encode languages other than English.

**(2)**

(b) Convert the denary number –43 to 8-bit binary using sign and magnitude representation.

**(2)**

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## 6. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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## 7. Question 3 · part c.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.i only.

**3** Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

**(1)**

.....

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

**(3)**

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(iii) One pixel in Image 1 is represented by the unsigned binary integer

1100 1010

Identify the denary equivalent.

**(1)**

☒ **A** 54

☒ **B** 191

☒ **C** 202

☒ **D** –74

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## 7. Question 3 · part c.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.i only.

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(b) A register holds the value 1100 0001 as an 8-bit binary value in two's complement.

(i) Convert the two's complement binary integer 1100 0001 to denary.

**(2)**

(ii) Give the result of applying an arithmetic shift left by one to the binary pattern 1100 0001

**(1)**

(iii) The value 1111 1111 is added to the value 1100 0001 stored in the register.

An error occurs.

Explain this error.

**(2)**

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## 7. Question 3 · part c.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.i only.

(c) The ASCII code for the character **1** is 011 0001

(i) Give the ASCII code for the character **5**

**(1)**

.....

(ii) Give **two** benefits of using ASCII to encode characters.

**(2)**

1 .....

.....

2 .....

.....

**(Total for Question 3 = 13 marks)**

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Working and answer

## 8. Question 3 · part c.ii · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part c.ii only.

**3** Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

**(1)**

.....

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

**(3)**

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(iii) One pixel in Image 1 is represented by the unsigned binary integer

1100 1010

Identify the denary equivalent.

**(1)**

☒ **A** 54

☒ **B** 191

☒ **C** 202

☒ **D** –74

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## 8. Question 3 · part c.ii · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part c.ii only.

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DO NOT WRITE IN THIS AREA

(b) A register holds the value 1100 0001 as an 8-bit binary value in two's complement.

(i) Convert the two's complement binary integer 1100 0001 to denary.

**(2)**

(ii) Give the result of applying an arithmetic shift left by one to the binary pattern 1100 0001

**(1)**

(iii) The value 1111 1111 is added to the value 1100 0001 stored in the register.

An error occurs.

Explain this error.

**(2)**

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# 8. Question 3 · part c.ii · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part c.ii only.

(c) The ASCII code for the character **1** is 011 0001

(i) Give the ASCII code for the character **5**

**(1)**

.....

(ii) Give **two** benefits of using ASCII to encode characters.

**(2)**

1 .....

.....

2 .....

.....

**(Total for Question 3 = 13 marks)**

DO NOT WRITE IN THIS AREA

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Working and answer

# Practice workbook · Run-length encoding (RLE)

Compressing a string or pixel grid with RLE, and explaining whether RLE suits given data.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-15

# 1. Question 1 · part b.ii · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part b.ii only.

**1** Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

**(2)**

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

**(2)**

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(iii) Convert the binary pattern 11000110 to hexadecimal.

**(2)**

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# 1. Question 1 · part b.ii · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

(b) Images are stored as binary data.

(i) A bitmap image is 400 pixels wide by 200 pixels high.

It has a colour depth of 12 bits.

Construct an expression to show how the image size, in bytes, is calculated.

Do **not** calculate the answer.

**(2)**

(ii) **Figure 1** is an image of an icon, set out in an 8 × 8 pixel grid.

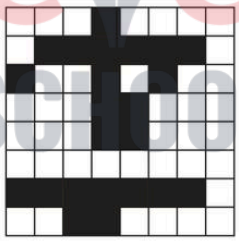


**Figure 1**

A run-length encoding compression algorithm uses:

- W for white pixels
- B for black pixels.

Give the result of compressing the first 4 lines of the icon using the algorithm.

**(3)**

Line 1 .....

Line 2 .....

Line 3 .....

Line 4 .....

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# 1. Question 1 · part b.ii · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part b.ii only.

(iii) Run-length encoding is a type of lossless compression.

State what is meant by the term **lossless** in this context.

(1)

(c) Raoul wants to record a song and store it as a digital sound file.

**Figure 2** shows an overview of the process of recording a song and storing it as a digital sound file.

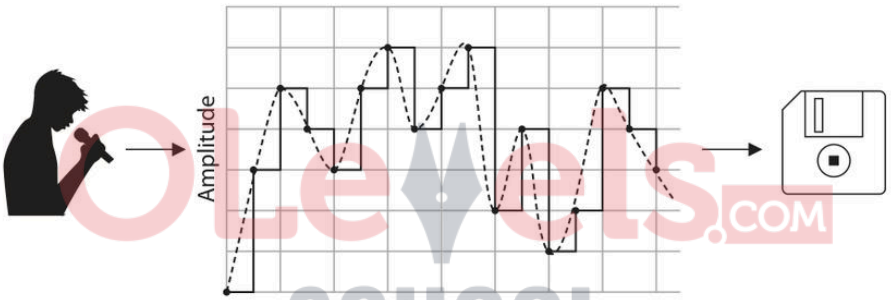



**Figure 2**

(i) Describe the steps taken to convert the analogue sound to a digital sound file.

(3)

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4

P 6 9 2 9 7 A 0 4 2 0

## 1. Question 1 · part b.ii · 3 marks · Working and answer

Working and answer

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# 2. Question 2 · part f.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f.iii only.

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**2** Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

**(1)**

(b) State the purpose of network protocols.

**(1)**

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

**(1)**

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

(ii) Give **one** task carried out by the network layer.

**(1)**

(d) Complete the table by adding a tick (✓) to match each item to its name.

**(3)**

| Item | Domain Name | IP Address | URL |
|---|---|---|---|
| https://www.pearson.com | | | |
| 216.33.251.195 | | | |
| pearson.com | | | |

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P 6 6 4 8 9 A 0 5 2 0

# 2. Question 2 · part f.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f.iii only.

(e) **Figure 2** shows a network topology.

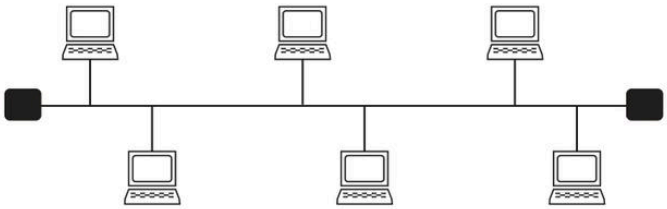


**Figure 2**

(i) State the name of this network topology.

**(1)**

(ii) State **one** benefit of using this network topology.

**(1)**

(iii) State **one** drawback of using this network topology.

**(1)**

(f) Files may be compressed using lossless or lossy algorithms.

(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

**(2)**

1 .....

2 .....

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# 2. Question 2 · part f.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f.iii only.

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(ii) Identify which **one** of these file formats uses lossy compression.

(1)

- ☒ **A** BMP
- ☒ **B** DOC
- ☒ **C** JPEG
- ☒ **D** PNG

(iii) Here is a string of data.

**CCCWWWCCWWWWWWCCC**

Give the result of compressing the string using a run-length encoding algorithm.

(1)

.....

.....

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(g) A video file is to be transmitted over the internet.

- The network transmission speed is 54 Mbps.
- The file size is 6 gigabytes (GB).

Construct an expression to show how the transmission time, in seconds, is calculated.

You do **not** have to do the calculation.

(4)

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**(Total for Question 2 = 18 marks)**

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## 2. Question 2 · part f.iii · 1 mark · Working and answer

Working and answer

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# 3. Question 3 · part e · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part e only.

**3** Software programs carry out tasks on computers.

(a) A headteacher uses a computer simulation to model the effect of increasing class sizes.

Explain **one** problem with using simulations to predict the effects of changes.

**(2)**

(b) Images are stored as files before being displayed or printed.

(i) Give **one** measurement of image resolution.

**(1)**

(ii) Increasing the colour depth of an image leads to an increase in the image file size.

Describe the reason for the increase in file size.

**(2)**

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8

P 7 2 9 3 7 A 0 8 2 4

## 3. Question 3 · part e · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part e only.

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(c) Software is divided into two categories.

Describe **one** difference between system software and application software.

**(2)**

(d) Here is an image of secondary storage.

Two files (W and Z) are stored on it.

Each file is made up of several blocks (e.g. Z1, Z2, Z3).

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | W4 | Z1 | | | | W2 | | W3 | Z3 | | W1 | | Z2 |

Complete the image to show the state after running a defragmentation utility.

**(2)**

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | |

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## 3. Question 3 · part e · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part e only.

(e) Here is part of a file that contains electric meter readings.

04631 04984 05103 05163 05271 05383 05487 05722

Explain the effect on the file of applying a run-length encoding algorithm to this data file.

**(2)**

.....

.....

.....

.....

(f) A series of tasks is written in different programming languages.

Complete the table to show the correct programming language translator for each task.

**(4)**

| Task | Translator |
|---|---|
| A guessing game that can be used on different computing platforms | |
| A screen driver for a new smartphone | |
| A new version of a spreadsheet program for sale next year | |
| Control software for an embedded system inside a new washing machine | |

**(Total for Question 3 = 15 marks)**

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## 3. Question 3 · part e · 2 marks · Working and answer

Working and answer

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# 4. Question 3 · part b.iii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.iii only.

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**3** Computers use binary to represent and store data.

(a) Binary and hexadecimal can be used to represent numbers.

(i) Convert the binary pattern 1011 1001 to hexadecimal.

**(2)**

(ii) Identify the statement that is true for hexadecimal.

**(1)**

- ☒ **A** Hexadecimal can be processed faster by a computer than binary
- ☒ **B** Hexadecimal digits can be represented using one nibble
- ☒ **C** Hexadecimal uses less storage space in RAM than binary
- ☒ **D** Hexadecimal uses the number base 6

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(iii) Describe the binary shift that is equivalent to dividing an unsigned binary number by 4.

**(2)**

# 4. Question 3 · part b.iii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.iii only.

(b) **Figure 1** shows a bitmap image in a 16 × 16 grid. The image uses three colours.

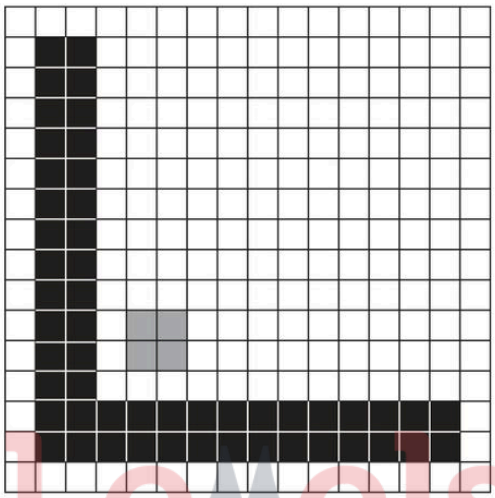


**Figure 1**

(i) State the smallest component of a bitmap image.

**(1)**

.....

.....

(ii) Construct an **expression** to show the number of images, like the one in **Figure 1**, that can be stored in 1 KB.

You do **not** need to do the calculation.

**(4)**

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# 4. Question 3 · part b.iii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.iii only.

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(iii) Explain why a run-length encoding algorithm is a good choice for compressing the image in **Figure 1**.

**(2)**

(c) The **Pigpen** cipher shown in **Figure 2** has been used to encrypt a message.

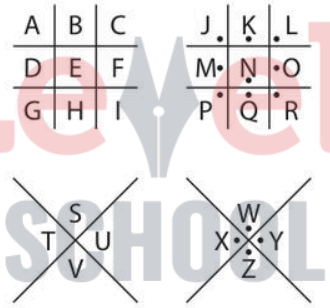
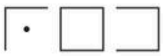



**Figure 2**

Complete the table to give the decrypted plain text.

**(2)**

| Encrypted text | Plain text |
|---|---|
|  | |
|  | |

**(Total for Question 3 = 14 marks)**

## 4. Question 3 · part b.iii · 2 marks · Working and answer

Working and answer

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# Practice workbook · Logical and arithmetic binary shifts

Logical and arithmetic shifts left or right by n places, and the multiply/divide effect of a shift.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-07

# 1. Question 2 · part c.ii · 1 mark

2023 JUN · Paper 01

Worked-example retry · Attempt part c.ii only.

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**2** Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

**(2)**

| Denary | Hexadecimal |
|---|---|
| 8 | |
| 12 | |

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

**(1)**

- ☒ **A** $6^2$
- ☒ **B** $6 \times 2$
- ☒ **C** $6^2 - 1$
- ☒ **D** $2^6$

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

**(1)**

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

**(1)**

P 7 2 9 3 7 A 0 5 2 4

# 1. Question 2 · part c.ii · 1 mark

2023 JUN · Paper 01

Worked-example retry · Attempt part c.ii only.

(d) Binary patterns can be interpreted as signed or unsigned integers.

(i) Convert the denary unsigned integer 60 to 8-bit binary.

(2)

(ii) Here is a binary bit pattern for a signed integer in sign and magnitude format.

1001 0110

Convert the binary bit pattern to denary.

Be sure to include a sign symbol in your answer.

(2)

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6

P 7 2 9 3 7 A 0 6 2 4

## 1. Question 2 · part c.ii · 1 mark · Working and answer

Working and answer

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## 2. Question 1 · part e · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part e only.

**1** Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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## 2. Question 1 · part e · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part e only.

(d) Complete the table by adding these two 8-bit binary integers.

**(2)**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| | | | | | | | |

(e) Give the result of performing a logical left shift of 1 place on the binary integer 0100.

**(1)**

(f) Give the number of bits per character used by standard ASCII.

**(1)**

(g) **Figure 1** shows a black and white bitmap image.

The pixels in row 5 are represented by the binary pattern 1001 1001

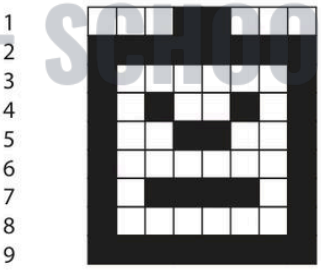



**Figure 1**

(i) State what is meant by the term **pixel**.

**(1)**

(ii) Construct an expression to show the number of pixels in the image.

**(1)**

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## 2. Question 1 · part e · 1 mark · Working and answer

Working and answer

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# 3. Question 2 · part c.i · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part c.i only.

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**2** Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

**(2)**

| Denary | Hexadecimal |
|---|---|
| 8 | |
| 12 | |

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

**(1)**

- ☒ **A** $6^2$
- ☒ **B** $6 \times 2$
- ☒ **C** $6^2 - 1$
- ☒ **D** $2^6$

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

**(1)**

.....

.....

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

**(1)**

.....

.....

P 7 2 9 3 7 A 0 5 2 4

5

*Turn over* ▶

## 3. Question 2 · part c.i · 1 mark · Working and answer

Working and answer

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# 4. Question 3 · part b.ii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b.ii only.

**3** Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

**(1)**

.....

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

**(3)**

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(iii) One pixel in Image 1 is represented by the unsigned binary integer

1100 1010

Identify the denary equivalent.

**(1)**

☒ **A** 54

☒ **B** 191

☒ **C** 202

☒ **D** –74

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# 4. Question 3 · part b.ii · 1 mark

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Further practice · Attempt part b.ii only.

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(b) A register holds the value 1100 0001 as an 8-bit binary value in two's complement.

(i) Convert the two's complement binary integer 1100 0001 to denary.

(2)

(ii) Give the result of applying an arithmetic shift left by one to the binary pattern 1100 0001

(1)

(iii) The value 1111 1111 is added to the value 1100 0001 stored in the register.

An error occurs.

Explain this error.

(2)

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## 4. Question 3 · part b.ii · 1 mark · Working and answer

Working and answer

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## 5. Question 3 · part a.iii · 2 marks

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Further practice · Attempt part a.iii only.

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**3** Computers use binary to represent and store data.

(a) Binary and hexadecimal can be used to represent numbers.

(i) Convert the binary pattern 1011 1001 to hexadecimal.

**(2)**

(ii) Identify the statement that is true for hexadecimal.

**(1)**

- ☒ **A** Hexadecimal can be processed faster by a computer than binary
- ☒ **B** Hexadecimal digits can be represented using one nibble
- ☒ **C** Hexadecimal uses less storage space in RAM than binary
- ☒ **D** Hexadecimal uses the number base 6

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(iii) Describe the binary shift that is equivalent to dividing an unsigned binary number by 4.

**(2)**

## 5. Question 3 · part a.iii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a.iii only.

(b) **Figure 1** shows a bitmap image in a 16 × 16 grid. The image uses three colours.

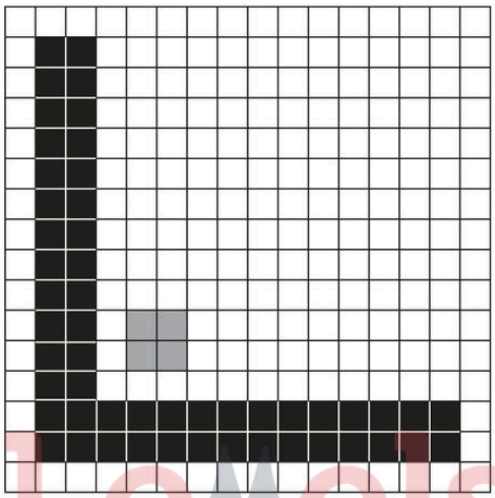


**Figure 1**

(i) State the smallest component of a bitmap image.

**(1)**

(ii) Construct an **expression** to show the number of images, like the one in **Figure 1**, that can be stored in 1 KB.

You do **not** need to do the calculation.

**(4)**

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## 5. Question 3 · part a.iii · 2 marks · Working and answer

Working and answer

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